

Importance of Preferential Flow and Soil Management in Vertical Transport of a Biocontrol Strain of *Pseudomonas fluorescens* in Structured Field Soil

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The large-scale release of wild-type or genetically modified bacteria into the environment for control of plant diseases or for bioremediation entails the potential risk of groundwater contamination by these microorganisms. For a model study on patterns of vertical transport of bacteria under field conditions, the biocontrol strain *Pseudomonas fluorescens* CHA0, marked with a spontaneous resistance to rifampin (CHA0-Rif), was applied to a grass-clover ley plot (rotation grassland) and a wheat plot. Immediately after bacterial application, heavy precipitation was simulated by sprinkling, over a period of 8 h, 40 mm of water containing the mobile tracer potassium bromide and the dye Brilliant Blue FCF to identify channels of preferential flow. One day later, a 150-cm-deep soil trench was dug and soil profiles were prepared. Soil samples were extracted at different depths of the profiles and analyzed for the number of CHA0-Rif cells and the concentration of bromide and Brilliant Blue FCF. Dye coverage in the soil profiles was estimated by image analysis. CHA0 was present at 10⁸ CFU/g in the surface soil, and 10⁶ to 10⁷ CFU/g of CHA0 was detected along macropores between 10 and 150 cm deep. Similarly, the concentration of the tracer bromide along the macropores remained at the same level below 20 cm deep. Dye coverage in lower soil layers was higher in the ley than in the wheat plot. In nonstained parts of the profiles, the number of CHA0-Rif cells was substantially smaller and the bromide concentration was below the detection limit in most samples. We conclude that after heavy rainfall, released bacteria are rapidly transported in large numbers through the channels of preferential flow to deeper soil layers. Under these conditions, the transport of CHA0-Rif is similar to that of the conservative tracer bromide and is affected by cultural practice.

The application of beneficial bacteria for biological control of plant diseases or bioremediation on a commercial scale requires the release of large numbers of wild-type or genetically modified strains into the environment. Concern about the ecological safety of such applications has been raised. Information on the behavior of these microorganisms after release, in particular on their persistence and possible dissemination to nontarget sites, is needed. The vertical transport of introduced microorganisms in field soil has been discussed for several reasons: (i) the transport of introduced organisms to groundwater could lead to contamination of drinking water or to horizontal transport in the aquifer to other nontarget sites (17, 18); (ii) the distribution of biocontrol bacteria along the root system is crucial for their performance (27, 36); and (iii) bioremediation is effective only if introduced strains spread to the contaminated sites (16).

Transport of bacteria has often been investigated in laboratory columns containing sand or homogenized soil (13, 19), and some mathematical models have been developed (32). On the basis of results from such experiments, it has been suggested that bacteria are effectively retained in soil by the mechanisms of straining in the soil matrix or adsorption on soil particles (25, 32). However, vertical translocation of bacteria is substantially increased in intact soil columns compared with repacked soil columns (26, 28, 34). Therefore, it has been proposed that macropore flow is much more important for

bacterial transport than matrix flow (28). Literature about vertical transport of microorganisms released for biocontrol or bioremediation under field conditions is scarce (2). Most of the available information on vertical transport of bacteria in the field comes from studies on the transport of fecal coliforms at wastewater disposal sites (2–4).

In field soil, the flow of water is often spatially irregular; water follows cracks and biopores and thereby bypasses the soil matrix. This phenomenon is called preferential flow (15). Preferential flow has been recognized as important for the transport of conservative tracers and pesticides in field soil during periods of high precipitation (20, 29, 37). Especially for colloids, preferential transport through macropores is important during periods of high infiltration (24). Bacteria are of colloidal size, and therefore concepts used in colloidal transport theory should also apply to bacterial transport (25).

The irregular flow patterns of water can be visualized in soil profiles by the addition of a dye to the irrigation solution (11, 14). Flury et al. (11) compared the susceptibility of different soils to preferential flow by using the anionic dye Brilliant Blue FCF. Flury and Flühler (10) described the tracer characteristics of this dye. One of the most commonly used tracers to monitor water movement in soil is bromide (12, 20). As a negatively charged, nonreactive anion, it does not adsorb to negatively charged soil constituents, and it can be quantified easily in soil samples.

Here we report field data about preferential transport of bacteria in soil with *Pseudomonas fluorescens* CHA0 as a model organism. Strain CHA0 is an effective biocontrol agent, which

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TABLE 1. Soil properties in the soil profiles of the ley plot and the wheat plot at the experimental site

Depth (cm)	pH	Content (% wt/wt) of:					
		Calcium	Organic C	Humic acids	Clay	Silt	Sand
Ley plot							
0-22	6.5	0	1.8	3.1	33.3	31.0	32.5
22-40	6.9	0	1.1	1.9	33.2	29.4	35.4
40-62	7.0	0	0.5	0.9	37.0	22.8	39.2
62-83	7.9	5	0.5	0.9	31.6	20.8	46.6
83-100	8.4	38	0.3	0.5	19.7	25.1	54.6
Wheat plot							
0-22	6.2	0	2.2	3.8	41.2	32.3	22.6
22-40	7.1	0	1.8	3.1	35.0	32.9	28.9
40-60	7.0	0	0.6	1.0	30.7	32.8	35.4
70-100	7.1	0	0.3	0.5	40.7	18.7	39.9
106-115	8.4	14	0.3	0.5	24.6	21.2	53.6

protects plants from a range of root diseases (5–7, 35). In previous studies, the vertical transport and survival of strain CHA0 have been monitored in intact soil columns under growth chamber conditions (26) and in large outdoor lysimeters (21, 33). The main purposes of this study were (i) to elucidate patterns of vertical movement of water and bacteria in field soil by using a colored sprinkling solution, (ii) to compare the vertical transport of bacteria with the transport of the nonreactive tracer bromide, and (iii) to determine the influence of soil management on bacterial transport by comparing a grass-clover ley (rotation grassland) plot with a wheat plot.

MATERIALS AND METHODS

Culture conditions and inoculum preparation. *P. fluorescens* CHA0 (31) marked with a spontaneous resistance to rifampin (CHA0-Rif [26]) was cultivated on King's medium B agar (22) containing 100 µg of rifampin per ml. For addition to soil, overnight cultures of CHA0-Rif grown on King's medium B agar at 27°C were harvested and suspended in distilled water. The suspension was filtered through three layers of cheesecloth, and its optical density at 600 nm was measured to adjust the cell density.

Description of the field site. The field site chosen for bacterial release represents a typical Swiss midland soil, a clayey loam (gleyic Cambisol). The soil is of alluvial origin with a coarse gravel layer below 0.8 to 1.0 m, overlaid with clayey and loamy layers. The soil profile shows some mottling properties with abundant manganese concretions. During rainfall, the soil is often waterlogged both below and above the plow pan. Table 1 describes some chemical and physical soil properties. Both plots have been under cultivation for about 20 years and show a rather compact soil structure, which in the ley plot (3-year-old rotation grassland with a grass-clover mixture) has become better structured as a result of drought-induced shrinkage, permanent root activity, and the presence of soil organisms, especially earthworms. Wheat was at the flowering stage and the volumetric water content was 43% when the trial was conducted.

Application of bacteria and chemical tracers to field plots. On 20 May 1994, vegetation on a 140-cm plot was cut to a height of 5 cm to allow uniform infiltration of the bacterial suspension and of the irrigation solution. One litre of a bacterial suspension containing 10^{10} CFU of CHA0-Rif per ml was applied per m² of soil surface with a sprayer. Immediately following the application of CHA0-Rif, 40 mm of an irrigation solution was applied to the field plots over a period of 8 h with an automatic sprinkler (11). This corresponds to a heavy rainstorm, which occurs once every 2 years in this region. Potassium bromide (0.015 mol/liter) was added to the irrigation solution as a mobile tracer. To identify channels of preferential flow, the irrigation solution contained 0.005 mol of Brilliant Blue FCF (*N*-ethyl-*N*-[4-[[4-ethyl(3-sulphophenyl)methyl]aminol]phenyl](2-sulphophenyl)-methylene]-2,5-cyclohexadien-1-ylidene)-3-sulphobenzene-methanaminium hydroxide inner salt, disodium salt; Plüss and Stauffer, Basel, Switzerland [commercial name, Vitasy Blue]; food-grade quality) per liter. Neither of the two tracers was found to be toxic to CHA0-Rif: 0.1 M KBr or 4 g of Brilliant Blue FCF per liter did not affect the growth of CHA0-Rif in liquid medium. The blue dye did not adsorb to CHA0-Rif cells, and it was not degraded by CHA0-Rif at a detectable rate within 72 h.

Preparation of soil profiles and sampling procedure. A 160-cm-deep trench was dug 1 day after application of CHA0-Rif to the surface soil of the field plots. Three subsequent 100-cm-wide by 150-cm-deep (ley) or 140-cm-deep (wheat

field) vertical soil profiles were then prepared at right angles to the original trench at a distance of 20 cm from each other and 50 cm away from the border of the sprinkled area. The profiles were finely prepared by breaking apart soil from the profile with sterilized knives; the knives were freshly sterilized for the preparation of each 5-cm soil layer in the profiles. The resulting surface of the profiles therefore never was in a direct contact with the tools used for digging and preparing of the profiles. To avoid contamination of samples taken from unstained soil matrix, another thin layer of soil was broken apart at the respective site on the profile immediately before sample extraction. A 10-cm by 10-cm grid was placed on each soil profile, and the blue dye pattern was photographed with a 35-mm camera with Kodachrome Elite 400 film for subsequent image analysis. Soil samples were extracted from the blue-stained (50 samples) and nonstained (10 samples) parts at different depths of each soil profile by pressing sterile steel cores (internal diameter, 2 cm) horizontally about 1.5 cm deep into the soil. In addition, a set of horizontal profiles (20 cm by 100 cm) were prepared between each of two subsequent vertical profiles at depths of 20, 40, 60, and 80 cm. From each horizontal profile, about 10 stained and 2 unstained samples were collected for analysis as described above.

Enumeration of bacteria and measurement of tracer concentration in soil samples. For monitoring the number of cultivable cells of CHA0-Rif, each soil sample (about 4 to 5 g) was transferred to a 25-ml flask containing 18 ml of sterile distilled water, shaken for 1 h at 300 rpm, and then vortexed vigorously for 15 s. Suspensions were decimally diluted and spread onto King's medium B agar containing 100 µg of rifampin per ml and 187.5 µg of actidione per ml. The plates were incubated at 27°C, and bacterial colonies that developed were counted after 3 days. The detection limit on this medium was about 50 CFU/g of soil. Bromide concentration was measured directly in the soil suspensions with an ion-selective electrode (Ingold Messtechnik, Urdorf, Switzerland). The detection limit was 1.5×10^{-8} mol per g of soil. For quantification of the Brilliant Blue FCF content in the soil samples, 2 ml of each soil suspension was centrifuged for 20 min at 14,000 rpm in an Eppendorf 5415C centrifuge and the optical density of the supernatant at 630 nm was determined. The detection limit for Brilliant Blue FCF is 10^{-9} mol per g of soil.

Image analysis. The blue-stained patterns on the photographs from the vertical and horizontal soil profiles were copied manually onto transparent paper. The resulting graphs were digitized by computer scanning with a resolution of three points per cm of soil profile. In vertical profiles, the percentage of dye coverage for every 1 cm of depth was calculated. For the horizontal profiles, the total percentage of dye coverage was calculated.

Statistics and estimation of mass recovery. The data obtained for bacterial populations and for the concentrations of the chemical tracers KBr and Brilliant Blue FCF approximated a logarithmic normal distribution. Therefore, means and standard deviations which are given in the figures were calculated with log-transformed values. In the vertical profiles, means of all samples per 10-cm-deep soil layer were calculated. Data given in the text are back-transformed logarithmic means.

For an estimation of mass recovery, the percentage of soil stained with Brilliant Blue FCF was multiplied by the total soil volume, the soil density, and the average number of bacteria or concentration of bromide tracer present in stained soil. The same calculation was performed with the percentage of unstained soil and respective bacterial numbers and tracer concentrations. This estimation is based on means of values without logarithmic transformation.

RESULTS

Identification and quantification of channels of preferential flow. To determine whether there is preferential flow of water in distinct macropores rather than a regular flow in the soil matrix, we added the dye Brilliant Blue FCF to the irrigation solution. The percentage of the soil profile covered with the dye then indicates the total volume of preferential flow paths. Therefore, the dye coverage in vertical (Fig. 1, panels A1 and B1) and horizontal (Fig. 2, panels A3 and B3) soil profiles of both the ley and the wheat plot was estimated by image analysis. In the ley plot, the dye coverage decreased drastically from 100% to less than 5% within the 10-cm-deep surface soil layer (Fig. 1, panel A1). In the soil layers below, the dye coverage increased and reached values of 10 to 20% between depths of 70 and 150 cm (Fig. 1, panel A1). The analysis of the horizontal profiles gave similar results: the dye coverage increased from 1.3% at a 20-cm depth to 9.2% at an 80-cm depth (Fig. 2, panel A3). In the wheat plot, the dye coverage was high (about 47%) in the top 20-cm soil layer but dropped to nearly 0% below the plough pan and then remained below 5% down to a 140-cm depth (Fig. 1, panel B1; Fig. 2, panel B3). This means that the water-conducting macropores were more com-

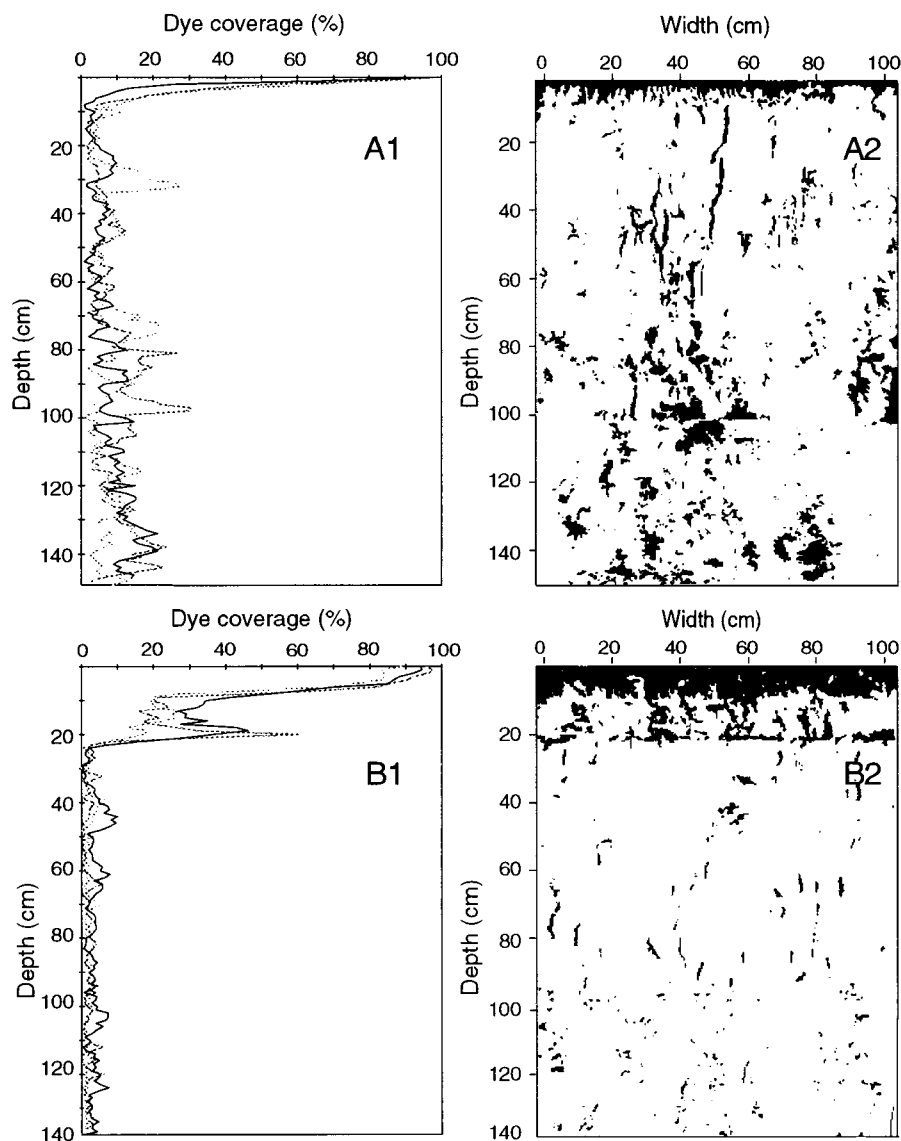


FIG. 1. Identification of water-conducting macropores in the soil profile of a ley plot (A1 and A2) and a wheat (B1 and B2) plot. (A1 and B1) Percentage of dye (Brilliant Blue FCF) coverage at different depths in three replicate soil profiles per plot estimated by image analysis. (A2 and B2) Representative soil profiles of each plot showing the distribution pattern of the dye and identifying channels of preferential flow. Profiles were excavated and photographed 24 h after application of 40 mm of a Brilliant Blue FCF dye solution containing the conservative tracer bromide. For details of the experimental setup, see Materials and Methods.

mon in the ley plot than in the wheat plot. A typical dye pattern in the vertical soil profile of the ley and wheat plots is shown in Fig. 1, panels A2 and B2. This pattern indicates that water flow was mainly along preferential flow paths. In the following, the term "macropores" will be used to refer to the stained areas in the soil profiles and the term "soil matrix" will refer to the unstained parts of the profile.

Number of CHA0-Rif cells and tracer concentration along macropores. At 1 day after bacterial application, the number of cultivable cells of CHA0-Rif in the completely stained 3-cm-deep surface soil layer of both plots was about 10^8 CFU/g of dry soil (Fig. 3, panels A1 and B1). Along the macropores in the subsurface soil between 10 and 150 cm deep, the cell numbers of CHA0-Rif ranged between 10^6 and 10^7 CFU/g of dry soil in both plots, regardless of the depth (Fig. 3, panels A1 and B1). The average numbers were 5.9×10^6 CFU/g in the ley plot and 2.5×10^6 CFU/g in the wheat plot. Analogous bac-

terial counts were found in soil samples extracted from stained areas in the horizontal profiles set up at different depths (Fig. 2, panels A1 and B1). These results suggest that after a heavy rainfall in the field soil, strain CHA0-Rif is transported in large numbers to deeper soil layers following the channels of preferential flow.

Potassium bromide had a similar pattern of vertical distribution in the soil profile to that of CHA0-Rif. The bromide concentration in the surface soil was about 5×10^{-6} mol/g of dry soil. Along the macropores between 20 and 150 cm deep, this concentration was about 4.5×10^{-7} mol/g in the ley plot, and 1.7×10^{-7} mol/g in the wheat plot (Fig. 3, panels A2 and B2). In the horizontal profiles set up at different depths, similar bromide concentrations were measured (Fig. 2, panels A2 and B2). The concentration of Brilliant Blue FCF in the surface soil of both plots was around 1.6×10^{-6} mol/g. In the macropores below 20 cm deep, the dye concentration was about

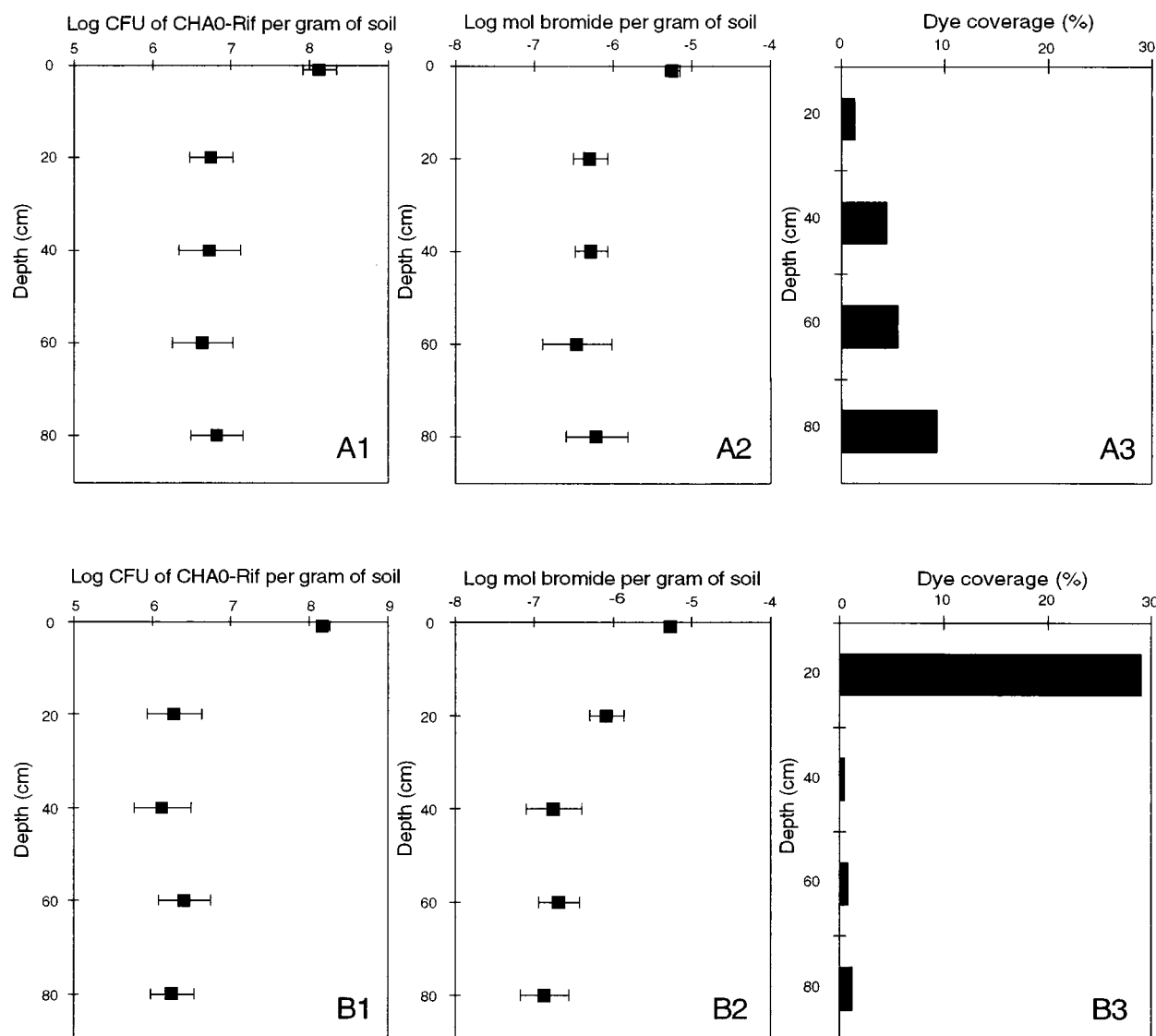


FIG. 2. CFU of *P. fluorescens* CHA0-Rif (A1 and B1) and concentration of the chemical tracer bromide (A2 and B2) along stained macro pores in horizontal soil profiles prepared at different depths in a ley plot (A1 to A3) and a wheat plot (B1 to B3). Means ($\pm$ standard deviations) were calculated from all log-transformed values from two (ley plot) or one (wheat field) horizontal profiles per depth. Dye coverage of Brilliant Blue FCF was estimated by image analysis (A3 and B3). Details of the experimental setup are described in the text.

2.5×10^{-7} mol/g in the ley plot and 7.1×10^{-8} mol/g in the wheat plot (data not shown).

Comparison of the transport of CHA0-Rif with the transport of bromide along macro pores. To compare the vertical transport of CHA0-Rif with the transport of bromide, the ratio of CFU of CHA0-Rif per mol of bromide was calculated. The ratio of the initially added bacterial inoculum to chemical tracer was 1.7×10^{13} CFU of CHA0-Rif per mol of bromide. At 1 day after application, the ratio was 2.7×10^{13} CFU/mol of bromide in the surface soil of both field plots. In the soil macro pores below a 20-cm depth, the ratio was constant (with little variation) at about 1.3×10^{13} CFU/mol of bromide in the ley plot and 1.0×10^{13} CFU/mol of bromide in the wheat plot, regardless of the depth in the profiles. This finding indicates that the retention of the bacterial cells and of the conservative tracer bromide was similar along the macro pores between 20 and 150 cm deep and that only in the surface soil layer were

bacteria filtered to a slightly higher degree than bromide. Comparison of the ratios of the number of CHA0-Rif cells per mol of Brilliant Blue FCF at different soil depths gave analogous results. The ratio of added bacteria to Brilliant Blue FCF was 5.0×10^{13} CFU/mol. At 1 day later, the ratio was 8.6×10^{13} CFU/mol in the surface soil; along the macro pores below a 20-cm depth, it was 2.1×10^{13} CFU/mol in the ley plot and 2.0×10^{13} CFU/mol in the wheat plot.

Number of CHA0-Rif cells and tracer concentration in the soil matrix. Strain CHA0-Rif was also detected in samples from the unstained soil matrix. In the ley plot, CHA0-Rif counts were above the detection limit (50 CFU/g of soil) for all except 2 of 52 soil samples and varied between 10^2 and 10^6 CFU/g of dry soil (Fig. 4A) with a mean of 10^5 CFU/g. In the wheat plot, the concentration of CHA0-Rif was below the detection limit in 9 of 38 samples and varied between 10^1 and 10^6 CFU/g in the other samples, with a mean of 2.5×10^4

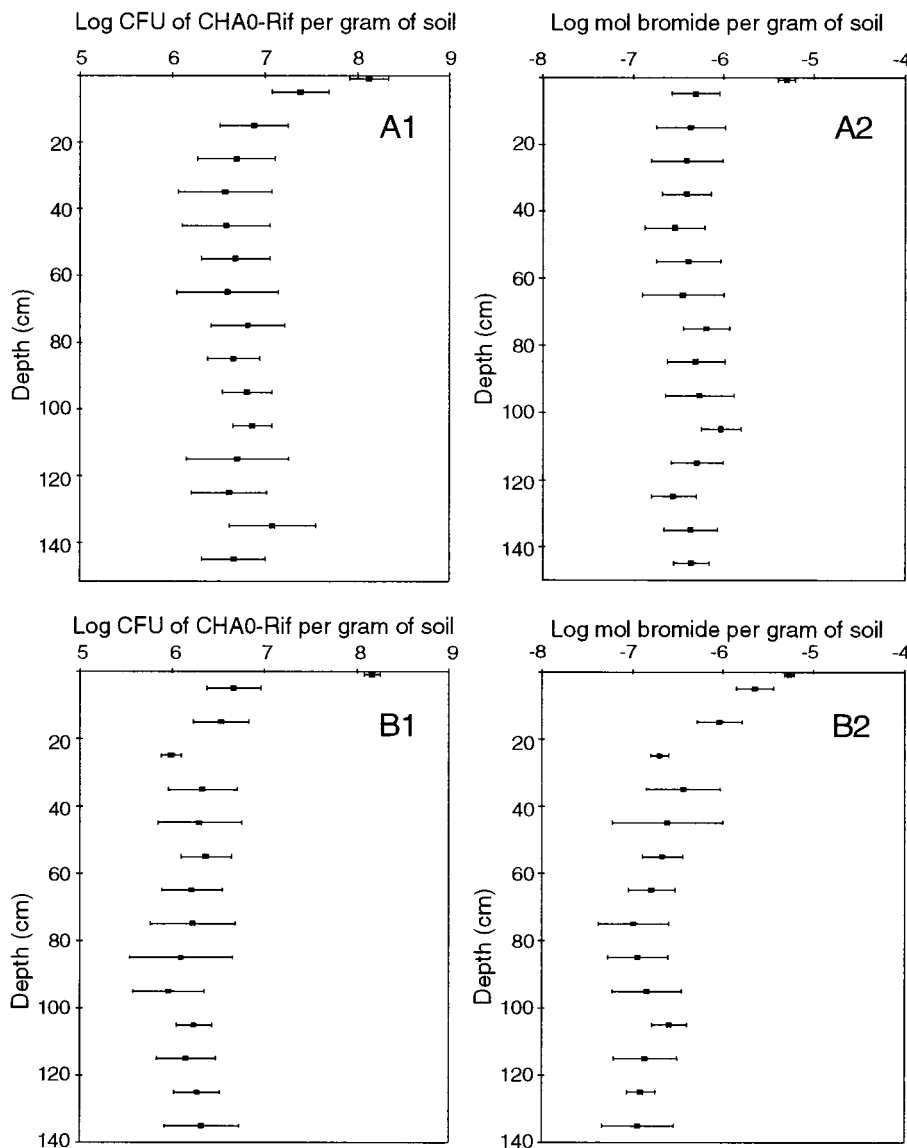


FIG. 3. Preferential transport of *P. fluorescens* CHA0-Rif and the conservative tracer bromide to deeper soil layers in a ley plot (A1 and A2) and a wheat plot (B1 and B2). Shown are CFU of CHA0-Rif (A1 and B1) and concentration of the chemical tracer bromide (A2 and B2) along stained soil macropores at different depths in vertical soil profiles; means ($\pm$ standard deviations) were calculated from all log-transformed values per 10-cm-deep soil layers from three subsequent profiles per plot. Soil samples were taken 24 h after application of 10^{13} CFU of CHA0-Rif per m^2 of surface. Immediately after application of the bacteria, heavy precipitation was simulated by sprinkling 40 mm of a dye solution containing 0.015 mol of potassium bromide per liter over a period of 8 h. For details of the experimental setup, see Materials and Methods.

CFU/g (Fig. 4B). In contrast, the concentration of the tracer bromide was above the detection limit (1.5×10^{-8} mol/g of dry soil) in only 5 of 52 samples from the ley plot and in 5 of 38 samples from the wheat plot.

Mass recovery. The estimation of mass distribution and mass recovery gives the percentage of applied CHA0-Rif cells and tracer which actually moved to deeper soil layers and reflects the efficiency of the recovery methods used in the experiments. The estimation is made on the basis of the following assumptions: (i) the samples extracted along the macropores are representative of the stained area quantified by image analysis; (ii) the three-dimensional proportion of stained soil corresponds to the two-dimensional proportion of the stained surface area of the vertical profiles recorded by image analysis; and (iii) cells of CHA0 neither multiplied nor died in the period be-

tween application of CHA0 and excavation of the soil profiles. The results obtained from this estimation are given in Table 2. About 55% of the bacteria and 33% of the applied bromide were retained in the 3-cm surface soil layer of both plots. In the ley plot, about 13% of the applied CHA0-Rif cells and 16% of the added bromide were detected in stained areas below a 25-cm depth, which represents about 9% of the soil volume in these layers. In the wheat plot, 1% of the applied CHA0-Rif cells and 2% of the bromide were detected in stained areas (about 2.5% of the soil volume) below a 25-cm depth (Table 2).

DISCUSSION

Results of this work suggest that in a situation with high precipitation immediately after release of bacteria into field

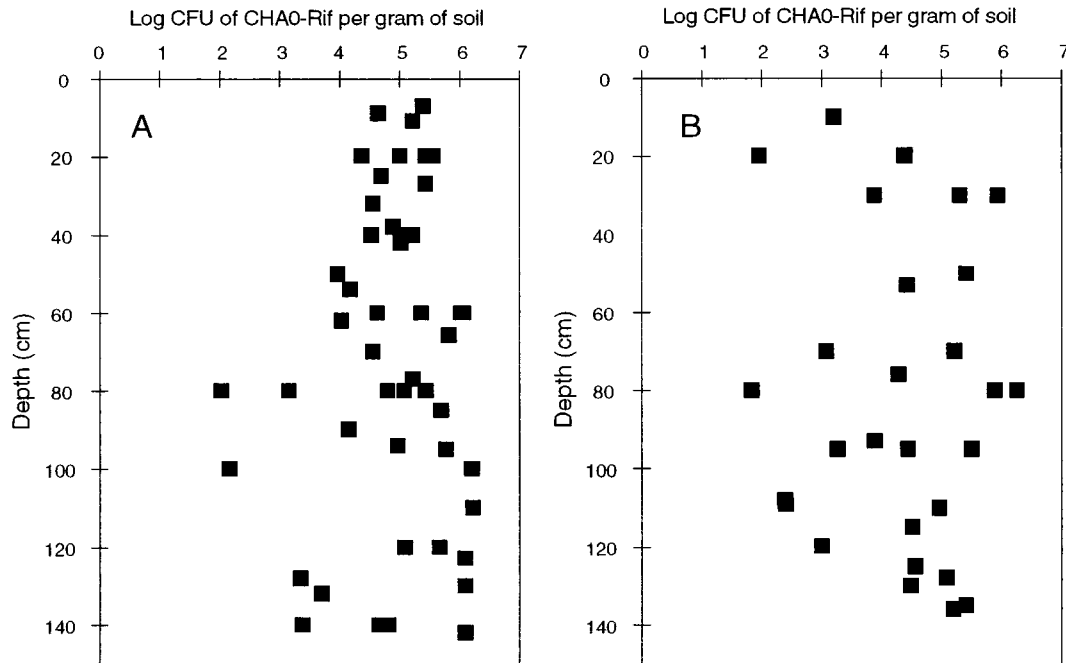


FIG. 4. Number of cultivable cells of *P. fluorescens* CHA0-Rif in soil samples extracted from unstained parts of vertical and horizontal profiles of a ley plot (A) and a wheat plot (B). Each point represents a single soil sample. CHA0-Rif counts were below the detection limit (50 CFU/g of soil) in 2 of 52 samples from the ley plot and 9 of 38 samples from the wheat plot. Details of the experimental setup are described in the text.

soil, macropore flow can lead to the displacement of a significant number of the applied bacteria to at least 150 cm deep. It is striking that the number of CHA0-Rif cells along the soil macropores was large and remained at a constant level from 20 to 150 cm deep. The patterns of dye distribution in the soil profiles clearly show that preferential flow occurred, and the higher concentration of CHA0-Rif cells and bromide tracer along macropores than in the soil matrix indicates that bacterial cells and tracer were transported by preferential flow. An additional experiment was conducted on a grassland plot at a distant field site with a similar soil type (our unpublished results). With the same inoculum and the same irrigation rate, very similar results were obtained, with counts of CHA0 at about 4×10^6 CFU/g along macropores. This shows that under

the same conditions, similar transport rates may occur at different field sites.

It is difficult to compare our data with other data on vertical transport of released bacteria in the field. Little is known about the transport of plant growth-promoting rhizobacteria to deeper soil layers. Drahos et al. (8) monitored vertical and lateral movement of released *Pseudomonas aureofaciens* cells in a field experiment without simulation of precipitation. They detected the introduced bacteria only above 40 cm deep. Most currently available information about vertical transport in field soil comes from experiments on coliform transport at wastewater disposal sites. Butler et al. (4) applied sewage effluents to a sandy loam over a period of 28 months. They found a 10^5 -fold reduction of *Escherichia coli* counts from the surface soil

TABLE 2. Estimated mass distribution and mass recovery of *P. fluorescens* CHA0-Rif and bromide 24 h after application to field plots^a

Application and recovery	Distribution in:							
	Ley plot				Wheat plot			
	CHA0-Rif		Bromide		CHA0-Rif		Bromide	
	CFU/m ²	%	mol/m ²	%	CFU/m ²	%	mol/m ²	%
Applied initially	1.0×10^{13}	100	0.600	100	1.0×10^{13}	100	0.600	100
Surface (0–3 cm)	5.5×10^{12}	55	0.190	32	5.6×10^{12}	56	0.202	34
Stained parts of profile (3–25 cm)	2.9×10^{11}	3	0.013	2	4.3×10^{11}	4	0.169	28
Stained parts of the profile (26–150 cm) ^b	1.3×10^{12}	13	0.094	16	9.0×10^{10}	1	0.009	2
Unstained soil matrix	7.8×10^{11}	8	BD ^c		2.1×10^{11}	2	BD	
Total mass recovery	7.9×10^{12}	79	0.297	50	6.4×10^{12}	64	0.380	63

^a Soil samples were taken 24 h after application of 10^{13} CFU of CHA0-Rif per m² of surface. Immediately after application of the bacteria, heavy precipitation was simulated by sprinkling 40 mm of a dye solution containing 0.015 mol of potassium bromide per liter over a period of 8 h. The percentage of dye (Brilliant Blue FCF) coverage at different depths in three replicate soil profiles per plot was estimated by image analysis. The three-dimensional proportion of stained soil was assumed to be the same as the two-dimensional proportion of stained surface of the soil profiles recorded by image analysis. Means of all samples within the corresponding soil layer were calculated with untransformed values of bromide concentration and bacterial counts.

^b 25 to 140 cm in the wheat plot.

^c BD, below detection limit.

layer down to 130 cm deep. Bouwer et al. (3) reported a 10^4 -fold reduction in coliform concentration in the top 60 cm of the soil profile. This would indicate that compared with the results of our study, vertical transport of bacteria is much more restricted, but the relevance of this wastewater experiment to a situation in which bacteria are released on a normal field site is questionable. Macropores may have been clogged after repeated sewage application, or earthworm channels may have been absent at these sites.

The transport of CHA0-Rif along macropores was similar to the transport of the conservative tracer bromide, since the ratio of CFU of CHA0-Rif per mole of bromide applied to the surface soil of the field plots was similar to the ratio observed along the macropores at different depths. The slightly higher ratio measured in 3-cm-deep surface soil indicates that only in this layer were bacteria retained more than was the tracer. Nonreactive tracers such as bromide do not adsorb to negatively charged soil and are assumed to move through soil similarly to water (12). Bacteria are usually negatively charged, but physical interactions with soil particles, especially hydrogen bonding, van der Waals forces, and hydrophobic attraction, have been described previously (reviewed in reference 30). Besides these rapid physical processes, slower biological adhesion processes may also restrict bacterial movement. Production of extracellular substances such as polymeric fibrils can lead to irreversible adhesion of bacteria to a substrate (23). Bacteria may migrate to protected sites (e.g., very small pores), where they are prevented from being washed out (30). Most of these biological processes are time and energy dependent. As Stotzky (30) pointed out, it is not yet established which of these mechanisms is of major importance for bacterial adhesion to soil.

Our results suggest that once macropore flow occurs and bacteria have entered the macropores with the moving water, adhesion processes do not play an important role, since the same fraction of the nonadsorbing bromide and the bacteria is deposited along macropores (Table 2). However, the experimental setup with simulated precipitation immediately after bacterial application represents an extreme case. In a situation with delay of the irrigation, slow adhesion processes could lead to irreversible deposition of a larger fraction of the bacteria in the surface soil layer. The irrigation volume used in this study represents a realistic but rare event. How often preferential flow occurs under conditions of natural rainfall will strongly influence possible groundwater contamination by released bacteria. Therefore, information on the occurrence of preferential flow seems to be of major importance for safety considerations relating to bacterial transport in soil.

The occurrence of preferential flow in certain soil types is determined mainly by the soil structure. Flury et al. (11) compared 14 different field sites and found that structured clayey soils are more susceptible to preferential flow than are nonstructured sandy soils. Besides abiotic factors, management practice and biotic factors may influence the number and structure of macropores (1, 9). In our study, image analysis of dye coverage revealed a difference between the ley plot and the wheat field. Higher dye coverage in the ley plot indicated that after growth of ley for several years, and without tillage, water-conducting macropores become more numerous. In addition, the structure of the macropores was recorded where soil samples were taken. In the ley plot, a larger fraction of the macropores was due to earthworms, while in the wheat plot, root channels were more frequent (data not shown).

Most puzzling is the relatively large number of CHA0-Rif cells found in many of the samples from the nonstained soil matrix, in which no bromide had been measured. On the av-

erage, bacterial counts in these samples were 1.7% (ley) or 0.85% (wheat) of the average values detected along stained macropores. On the one hand, this could be attributed to the much more sensitive detection method for bacteria than for bromide. On the other hand, in samples from the soil matrix with more than 2×10^5 CFU of CHA0-Rif per g and no detectable bromide, the ratio of CFU of CHA0 to moles of bromide must be higher than along the macropores, and some unknown mechanisms must have led to relatively greater bacterial numbers. As the preirrigation moisture was relatively high and most unstained parts are within 2 to 4 cm of stained macropores, active motility from stained parts through macropores to unstained parts is feasible, but bacterial motility in soil with native structure has not yet been investigated. *P. fluorescens* has been shown to move 10 to 26 mm within 24 h in disturbed soil, depending on nutrient availability (38). Harvey et al. (18) reported transport of bacteria through a sandy aquifer which was faster than that of a bromide tracer, and they attributed this to a size exclusion or hydrodynamic chromatography effect.

In conclusion, the results of this study suggest that a heavy rainfall occurring after application of bacteria to field soil leads to the transport of a significant number of the bacteria to depths of at least 150 cm. This transport is similar to that of a nonreactive tracer and is affected by agricultural practice.

The large number of introduced bacteria present in deeper soil layers after heavy precipitation raises the question about bacterial survival in these layers, which might represent a site for long-term survival of introduced strains. Further studies to answer these questions are in progress.

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